

Project Ideation & Architectural Decisions

1. Technology Stack Ideation

We selected Python and Flask for the core server architecture to allow native integration with Scikit-learn, joblib, and pandas. Flask's lightweight WSGI routing enables extremely fast REST API interfaces (<10ms routing latency).

For the frontend, we chose Vanilla HTML, CSS, and JS rather than complex frameworks to keep the project light, accessible, and fast, and styled it with custom CSS variables, Glassmorphism panels, and smooth transition animations.

For storage, SQLite3 was selected due to its lightweight, serverless design, making the project immediately portable without complex external setups.

2. Machine Learning Algorithm Strategy

We brainstormed and benchmarked 5 algorithms (Logistic Regression, Decision Trees, Random Forests, Support Vector Machines, K-Nearest Neighbors).

Support Vector Machine (RBF Kernel) was chosen as the primary classifier because it maximizes category separation boundaries in high-dimensional spaces. Random Forest Regressor serves as the score estimator. Inputs are log-transformed on GNI and centered via standard scaler to ensure model alignment.